

Total No. of Questions : 10]

SEAT No. :

P3113

[Total No. of Pages : 4

[5461]-150-A

B.E. (Mechanical)

FINITE ELEMENT ANALYSIS - II

(2012 Course) (Semester-II) (Elective-IV) (402050 B)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Figures to the right indicate full marks.
- 2) Use of electronic pocket calculator is allowed.
- 3) Assume suitable data if necessary.

Q1) a) With neat sketch, explain p and h methods of mesh refinement in Finite Element Analysis. [6]

b) State and explain characteristics of global stiffness matrix. [4]

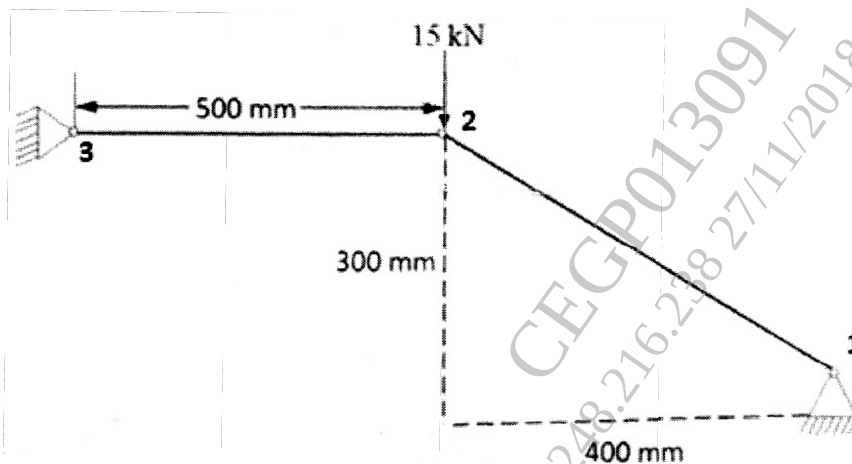
OR

Q2) a) Explain “Galerkin Weighted Residual Method” to formulate FEM equations. [6]

b) State stress-strain relations for plane stress and plane strain formulations in elasticity. [4]

Q3) a) Derive the expression for stiffness matrix for bar element using classical approach. [4]

b) For the two-bar truss shown in figure below, determine the displacement of node 2 and stresses in the element. All the elements have $E = 70 \text{ GPa}$ and $A = 200 \text{ mm}^2$. [6]



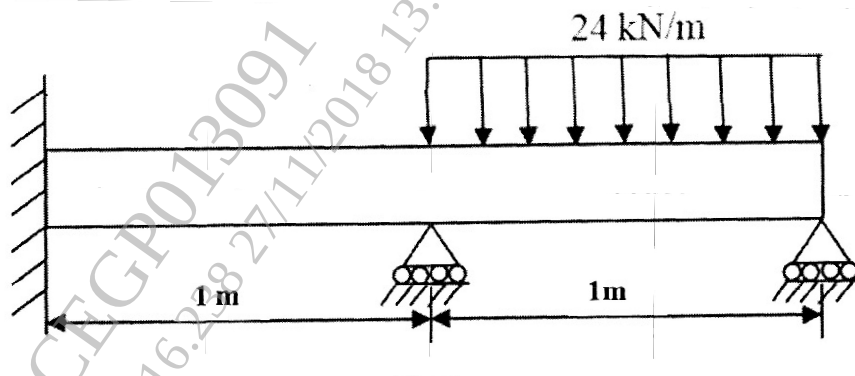
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P.T.O.

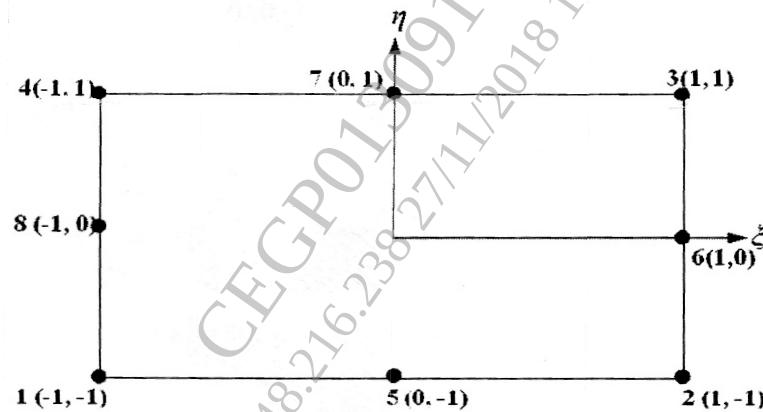
Q4) For the beam shown in figure below, determine:

[10]

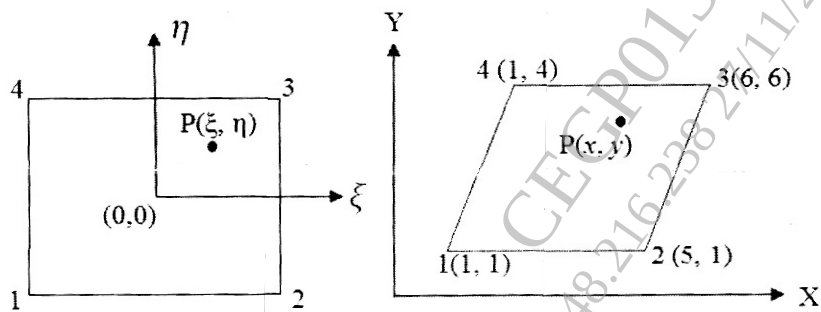
- Slopes at each node and,
 - Vertical deflection at the mid-point of the distributed load.
- Take $E = 200 \text{ GPa}$ and $I = 5 \times 10^6 \text{ mm}^4$



Q5) a) Obtain the shape functions N_1, N_2, N_3 and N_7 for following serendipity eight node rectangular element. [8]



- For four node quadrilateral element, find the x and y coordinates of point P whose location in parent element are given by. Also find u, v displacements of point P in X and Y directions respectively if displacement vector is $\{q\} = [0, 0, 0.20, 0, 0.15, 0, 0.10, 0, 0.05]^T$. [10]



OR
2

- Q6)** a) What is full and reduced integration in Finite Element Analysis? [4]
 b) Write a short note on Patch Test. [4]
 c) Evaluate $I =$ using Gauss Quadrature method. [10]

- Q7)** a) Derive elemental stiffness matrix (conduction + convection) formulations for 1D steady state heat transfer problems. [8]
 b) A metallic fin, with thermal conductivity $70 \text{ W/m } ^\circ\text{K}$, 1 cm radius and 5 cm long extends from a plane wall whose temperature is 140°C . Determine the temperature distribution along the fin if heat is transferred to ambient air at 20°C with heat transfer coefficient of $5 \text{ W/m}^2 \text{ } ^\circ\text{K}$. Take two elements along the fin. Assume that the tip of fin is insulated. [8]

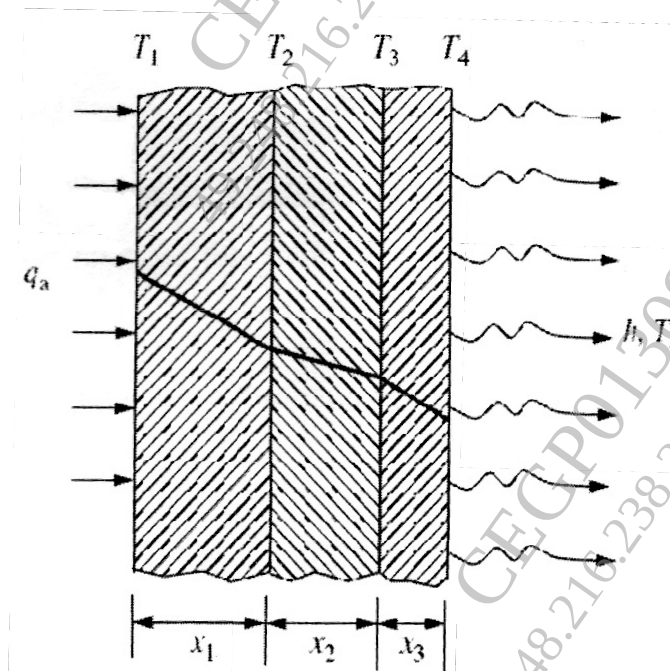
OR

- Q8)** A composite wall, with three layers of different materials as shown in figure below has the following properties for the different layers: [16]

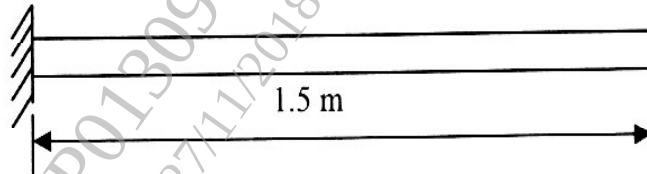
Layer-1: Gypsum, $k_3 = 0.05 \text{ W/m } ^\circ\text{C}$, $x_3 = 1 \text{ cm}$ and $q = 15 \text{ W/m}^2$

Layer-2: Fibre-glass, $k_2 = 0.0332 \text{ W/m } ^\circ\text{C}$ and $x_2 = 5 \text{ cm}$

Layer-3: Concrete, $k_1 = 1.2 \text{ W/m } ^\circ\text{C}$, $x_1 = 15 \text{ cm}$, $h = 15 \text{ W/m}^2 \text{ } ^\circ\text{C}$ and $T_a = 25^\circ\text{C}$. Calculate the temperatures T_1 , T_2 , T_3 and T_4 assuming unit area of heat flow.



- Q9)** Estimate natural frequencies of axial vibrations of bar shown in figure below, using both consistent and lumped mass matrices and compare the results. Bar is having uniform cross-section with cross-sectional area $A = 20 \times 10^{-6} \text{ m}^2$, length $L = 1.5 \text{ m}$, modulus of elasticity $E = 2 \times 10^{11} \text{ N/m}^2$ and density $= 7800 \text{ kg/m}^3$. Model the bar by using two elements. [16]



OR

- Q10)a)** Derive consistent mass matrix for bar, beam and truss element. [12]
- b) Explain the significance of lumped mass matrix and consistent mass matrix. Write lumped mass matrix for bar and beam element. [4]

